# Soil DNA extraction: SuperSoil, manual (legacy; current method = semi-automated Zixpress 32)

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**Abstract**

A method for fungal and bacterial DNA extraction.

**Safety warnings**

1. Solution SEA and Ethanol are flammable
2. DO NOT add bleach or acidic solutions directly to the sample preparation waste.

**Materials**

1. Bead (ball mill)

| **Zirconia beads** | **0.5** | **6** | **0.2** | **CAAR-ZB-0.5-1000BX** |
|---|---|---|---|---|
| Zirconia beads | 1.0 | 6 | 0.2 | CAAR-ZB-1-1000BX |

2. Solution SD1 (Chaotropic)

| | **w/w** |
|---|---|
| sodium thiocyanate | 2.5% |
| $Na_2HPO_4$ | 5% |
| pH | ~8.4 |

3. Solution SD2

| | **w/w** |
|---|---|
| $AlCl_3 \cdot 6H_2O$ | 1.5% |
| Ammonium acetate | 25% |
| pH (Adjust with acetic acid) | 6.3 |

4. Solution SD3 (Chaotropic)

| | **w/w** |
|---|---|
| Guanidinium thiocyanate | 65% |
| pH | ~7.0 |

Solution SEA (Wash buffer 1)

| | **w/w** |
|---|---|
| Ethanol | 50% |
| Isopropanol | 10% |
| Guanidine hydrochloride | 30% |

80% Ethanol (Wash buffer 2)

DEPC treated water (Elution buffer)

**Procedure**

**Sample preparation & Cell lysis**

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**1.** Add **0.2 mL** ⌀=1.0 mm zirconia beads, **0.2 mL** ⌀=0.5 mm zirconia beads, **800 µL** SD1 solution, **0.5 g** sample into a 1.5 ml microcentrifuge tube. Vortex briefly to mix.

※ **Note:** If the SD1 solution is precipitated, leave it at room temperature for 30 minutes, it will become transparent again.

※ **Note:** If the soil is too loose and fills up the whole microcentrifuge tube, you can separate them into 2 microcentrifuge tubes for the downstream workflow. And centrifuge 2 microcentrifuge tubes at the final step to merge them.

※ **Note:** If your sample is hard to lyse, you can try the following methods:

1. After adding solution SD1, incubate at 65 °C for 10 minutes
2. Add 3 additional stainless steel beads and use 2.0 ml screw vials for your samples

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**2.** (4 min) Homogenize samples using Vortex-Genie® 2 with SI-H524 Horizontal microtube holder. Vortex at maximum speed for **4 min**

※ **Note:** Be careful when removing the microcentrifuge tube from the holder. Press the lid and slowly loosen the tube with another hand.

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**3.** (2 min) Centrifuge the 1.5 ml microcentrifuge tube at **14000 rpm, 25 °C, 2 min**

**Inhibitor precipitation**

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**4.** Add **200 µL** SD2 solution to a clean 1.5 ml microcentrifuge tube.

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**5.** (5 s) Transfer **500 µL** supernatant to the tube in the previous step. Vortex **5 s** for mixing.

※ **Note:** White precipitate forms.

If too much AlCl3 (CD2) is added, it will cause all the DNA to be precipitated along with the inhibitory substances.

※ **Note:** If there is any suspended substance, you can use P200 and turn it to 150 µl and suck the supernatant up 3 times.

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**6.** (2 min) Centrifuge at **14000 rpm, 25 °C, 2 min**

**DNA binding**

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**7.** (5 s) Add **600 µL** SD3 solution to a clean 1.5 ml microcentrifuge tube. Avoiding the pellet, transfer up to **700 µL** of supernatant from the previous step to that tube. Vortex **5 s** for mixing.

※ **Note:** If Solution CD3 has precipitated, heat at 60°C until the precipitate dissolves.

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**8.** (2 min) Take **650 µL** to spin column (with collection tube) and centrifuge at **14000 rpm, 25 °C, 2**

**min**

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**9.** Discard the flow-through and repeat step 8 to ensure all the lysates have passed through the spin column.

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**10.** Carefully place the spin column into a clean 2 ml collection tube. Avoid splashing any flow-through onto the spin column.

**Wash**

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**11.** (2 min) Add **500 µL** of Solution SEA to the spin column. Centrifuge at **14000 rpm, 25 °C, 2 min**

※ **Note:** Solution SEA may leaks or suck back causing contamination.

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**12.** Discard the flow-through and return the spin column to the same 2 ml collection tube.

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**13.** (2 min) Add **500 µL** 80% ethanol to the spin column. Centrifuge at **14000 rpm, 25 °C, 2 min**

※ **Note:** Ethanol may leaks or suck back causing contamination.

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**14.** Discard the flow-through and place the spin column into a new 2 ml collection tube.

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**15.** (3 min) Centrifuge at **14000 rpm, 25 °C, 3 min** Carefully place the spin column into the new 1.5 ml microcentrifuge tube

※ **Note:** Ethanol can interfere with downstream DNA applications, such as PCR, restriction digests, and gel electrophoresis. Make sure to centrifuge for 3 minutes.

**Elute**

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**16.** Add **50 µL** RNase-free water (DEPC-treated water) to the center of the white filter membrane.

※ **Note:** Alternative for RNase-free water: TE(Tris-EDTA) buffer

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**17.** (2 min) Centrifuge at **14000 rpm, 25 °C, 2 min** Discard the spin column. The DNA will be in the microcentrifuge tube.

**Quality and quantity assessment**

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**18.** Use BioTek Take3 & Take3 Trio Microvolume Plates to measure the quality and quantity of your specimens. And then upload them to Google Drive (1 MolecularWork (Direct edit) 分生實驗號碼(直接編輯) w_ Primer list)